

1 Implementation Guide TRT

This section describes how HYDRA-TRT (Tracking and Tracing) interacts with DMC (Dynamic Manufacturing Control). The document is based on the DMC Implementation Guide.

Further applicable documents

The following documents contain basic technical information on DMC or add up to the contents of this document:

- DMC Implementation Guide
- DMC Implementation Guide MDE
- DMC-SDK (included in DMC delivery)
 - o DMC-API documentation
 - o Sample ImplementationGuide

The SDK Sample ImplementationGuide provides examples on how TRT integrates in the reference process.

Tracking and Tracing

Configure DMC as follows in order to enable the functions and to integrate them using HYDRA.

Include the component *MaterialManager* into the FactoryModel at each station. Assign the component to the corresponding workplace as child.

```
<ComponentConfiguration Name="MaterialManager"
    Class="mpdv.MachineDataCollector.Dmc.Material.MaterialManager" />
```

This component can be supplied with material scan information via the inputs of the MaterialManagerCapability. Refer to the DMC API documentation for further details on the component and the corresponding Capability.

The recorded material information is stored in the workpiece. If you want to send this information to HYDRA, you must use the component MaterialMovementRepository.

```
<ComponentConfiguration Name="MaterialMovementRepository"
    Class="mpdv.MachineDataCollector.Dmc.Material.MaterialMovementRepository">
    <Parameter Name="WriteConnector" Value="JtpPersistentConnector" />
</ComponentConfiguration>
```

In this example, you have configured the use of a safe connection to HYDRA via the parameter *WriteConnector*.

You use the component *MaterialAssignmentManager* to check if a material can be installed in an item. This component implements the interface *IMaterialAssignmentManager*:

```
string TryAssign(Workpiece workpiece, Material inputMaterial, Material candidate);
```

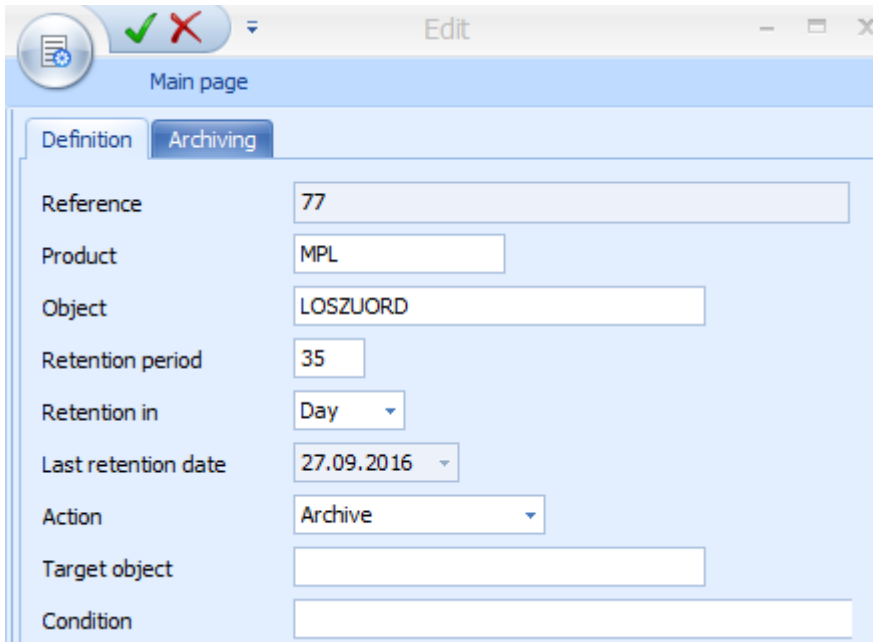
The *MaterialManager* then asks if the assignment is possible with the component that implements the interface. If you implement this interface, you can e.g. create a component that stores the installed serial numbers in a database. This component then ensures that the numbers are unique.

Configuration of TRT archiving

Contrary to the default archiving, the trace data recorded in DMC is not batch-related. The trace data is immediately archived if no batch is assigned to the trace data. This condition is always true for DMC data. For this reason, you must change the TRT default archiving.

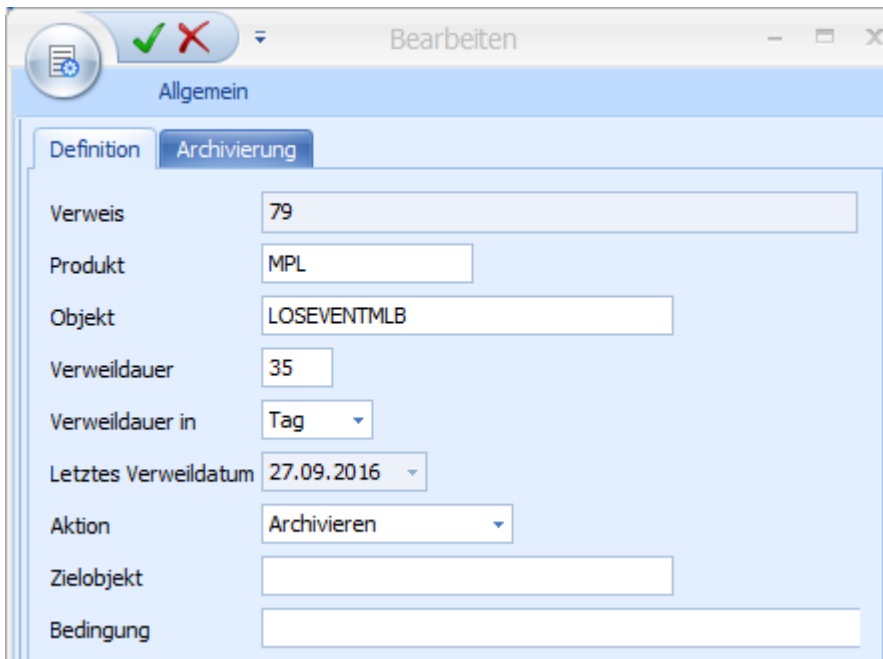
Delete the condition for the object LOSZUORD and set the retention period to e.g. 35 days:

Allgemein	
Definition	Archivierung
Verweis	77
Produkt	MPL
Objekt	LOSZUORD
Verweildauer	35
Verweildauer in	Tag
Letztes Verweildatum	27.09.2016
Aktion	Archivieren
Zielobjekt	
Bedingung	



Reference	77
Product	MPL
Object	LOSZUORD
Retention period	35
Retention in	Day
Last retention date	27.09.2016
Action	Archive
Target object	
Condition	

Delete the condition for the object LOSEVENTMLB and set the retention period to e.g. 35 days:



Verweis	79
Produkt	MPL
Objekt	LOSEVENTMLB
Verweildauer	35
Verweildauer in	Tag
Letztes Verweildatum	27.09.2016
Aktion	Archivieren
Zielobjekt	
Bedingung	


✓
✗
Edit

Main page

Definition
Archiving

Reference	79
Product	MPL
Object	LOSEVENTMLB
Retention period	35
Retention in	Day
Last retention date	27.09.2016
Action	Archive
Target object	
Condition	